

Project Title: Subreddit Search System Using Binary Search Trees

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Project Idea

The goal of this project is to build a simple application that organizes and searches real-world data using a Binary Search Tree. The application will load a dataset containing thousands of subreddit records and allow users to search for a specific subreddit quickly.

Each subreddit will be stored as a node in the BST using the subreddit name as the key while other information such as subscriber count or description will be stored as metadata. This will allow the system to efficiently organize and search the data.

Dataset

For this project we will use a public dataset of Reddit subreddits available online. The dataset includes several fields like subreddit name number of subscribers description creation date and category or topic. We will work with a dataset containing at least 10,000 entries to meet the scale requirement of the assignment. Because the data comes from real online sources it may contain inconsistencies such as different text formats missing values or special characters. Therefore before inserting the data into the Binary Search Tree (BST) we will perform preprocessing to clean and standardize the dataset. This includes converting subreddit names to

lowercase, removing unnecessary special characters, handling missing values in metadata fields, and removing duplicate entries if necessary. These preprocessing steps help ensure that the data can be stored properly and searched efficiently using the BST

Algorithm

The main algorithm used in this project will be a Binary Search Tree (BST).

Each node in the tree will represent a subreddit. The subreddit name will act as the key and the other dataset fields will be stored as additional information.

The BST will support the following operations:

- Insert subreddit records into the tree
- Search for a subreddit by name
- Traverse the tree to display sorted subreddit names

Using a BST allows efficient searching and insertion operations typically with $O(\log n)$ average time complexity.

Expected Outcome

By the end of the project we will have an application that can load more than 10,000 subreddit records and organize them using a Binary Search Tree. Users will be able to search for subreddit names quickly and view sorted results.

This project demonstrates how a Binary Search Tree can be applied to real-world datasets and how preprocessing is important when working with noisy data.